Integrated Cheminformatics Platform for Drug-Likeness Evaluation and Drug–Drug Interaction Analysis

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ABSTRACT:

This study presents the development of a computational platform for drug-likeness evaluation and drug–drug interaction analysis using cheminformatics approaches. A web-based application was designed to calculate key molecular descriptors, including molecular weight, lipophilicity (LogP), hydrogen bond donors and acceptors, and topological polar surface area (TPSA). Drug-likeness was assessed using Lipinski's Rule of Five.

The platform was applied to analyze interactions between commonly used drugs, namely Ibuprofen, Aspirin, and Paracetamol. The interaction potential was evaluated based on physicochemical properties and pharmacological mechanisms. The combination of Ibuprofen and Aspirin demonstrated a higher interaction risk due to overlapping targets and similar activity profiles, whereas Paracetamol and Ibuprofen showed lower interaction potential.

The results obtained were validated against established databases such as PubChem and SwissADME, confirming consistency and reliability. This work demonstrates how computational tools can be used for preliminary prediction of drug compatibility and supports the application of cheminformatics in medicinal chemistry.

INTRODUCTION:

The use of cheminformatics in medicinal chemistry has greatly improved the ability to analyze molecular structures and predict drug behavior with computational tools. Molecular descriptors like lipophilicity, polarity, and hydrogen bonding capacity are crucial for determining the pharmacokinetic and pharmacodynamic properties of drug molecules.

Lipinski's Rule of Five is commonly used to assess drug-likeness. It helps identify compounds with good absorption and bioavailability. Besides evaluating individual drugs, understanding drug-drug interactions is important for ensuring safety and effectiveness in treatment applications.

Drug-drug interactions can occur due to similar molecular properties, competition for biological targets, or shared metabolic pathways. Analyzing these interactions using computational methods can help identify potential risks before clinical trials begin.

This study aims to combine molecular descriptor analysis with interaction prediction by developing a web-based cheminformatics platform and validating it against established chemical databases.

OBJECTIVES:

- Develop a computational platform for molecular descriptor analysis using cheminformatics tools.
- Evaluate drug-likeness based on Lipinski's Rule of Five.
- Analyze drug-drug interactions using physicochemical and pharmacological parameters.
- Validate computational results with PubChem and SwissADME databases.
- Interpret the relationship between molecular descriptors and interaction potential.

METHODOLOGY:

The study was conducted using a computational cheminformatics workflow integrating data retrieval, molecular descriptor calculation, and interaction analysis. A web-based application was developed to serve as the primary analysis platform.

Drug data were obtained from PubChem, including molecular structure, molecular formula, and SMILES notation. The SMILES representation of each drug was used as input for molecular analysis. The computational backend of the application was implemented using RDKit, which enabled the calculation of key molecular descriptors such as molecular weight (MW), lipophilicity (LogP), hydrogen bond donors (HBD), hydrogen bond acceptors (HBA), and topological polar surface area (TPSA).

Drug-likeness was evaluated based on Lipinski's Rule of Five, which provides criteria for assessing the suitability of compounds for oral bioavailability. Three commonly used drugs—Ibuprofen, Aspirin, and Paracetamol—were selected for analysis based on their widespread clinical use and differing physicochemical properties.

Drug–drug interaction analysis was performed by comparing molecular descriptors and evaluating pharmacological mechanisms. Particular emphasis was placed on identifying similarities in target enzymes and physicochemical properties that may contribute to interaction potential. Combinations of Ibuprofen–Aspirin and Paracetamol–Ibuprofen were analyzed to represent high-risk and low-risk interaction scenarios, respectively.

To ensure reliability, the results obtained from the developed web application were cross-validated with established cheminformatics resources, including SwissADME for descriptor analysis and PubChem for structural and molecular data. Consistency between the computed results and reference databases was used as a measure of accuracy.

All analyses were performed under standard computational conditions without experimental or clinical intervention, focusing on theoretical and predictive evaluation within the scope of cheminformatics and medicinal chemistry.

User Interface and Molecular Visualization Output

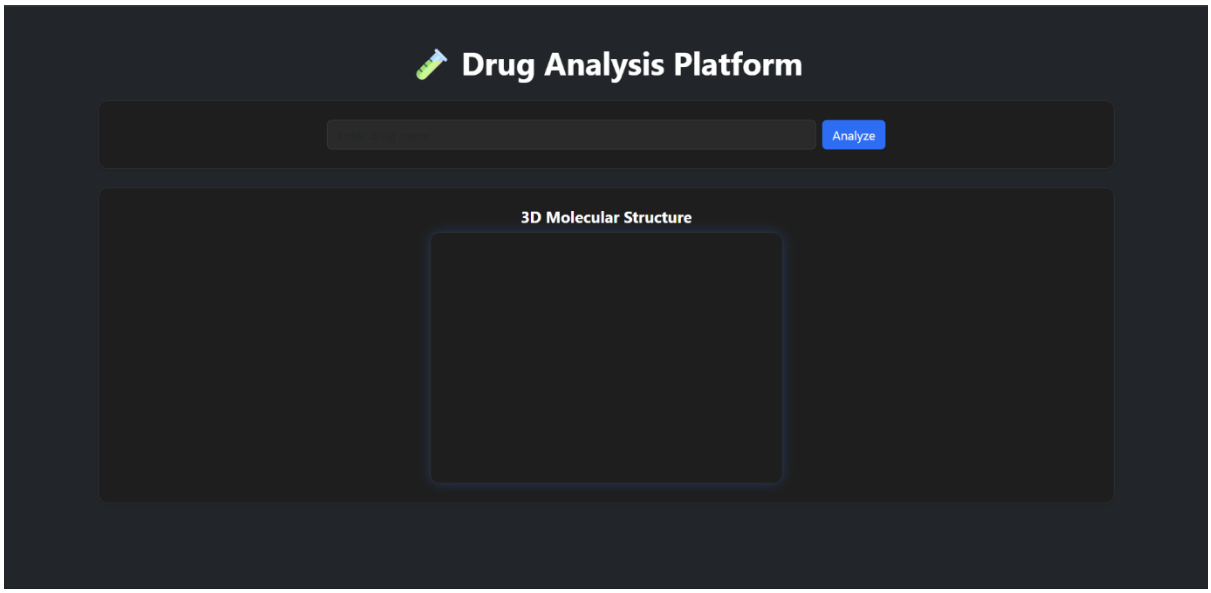


Figure 1: Developed web-based drug analysis platform

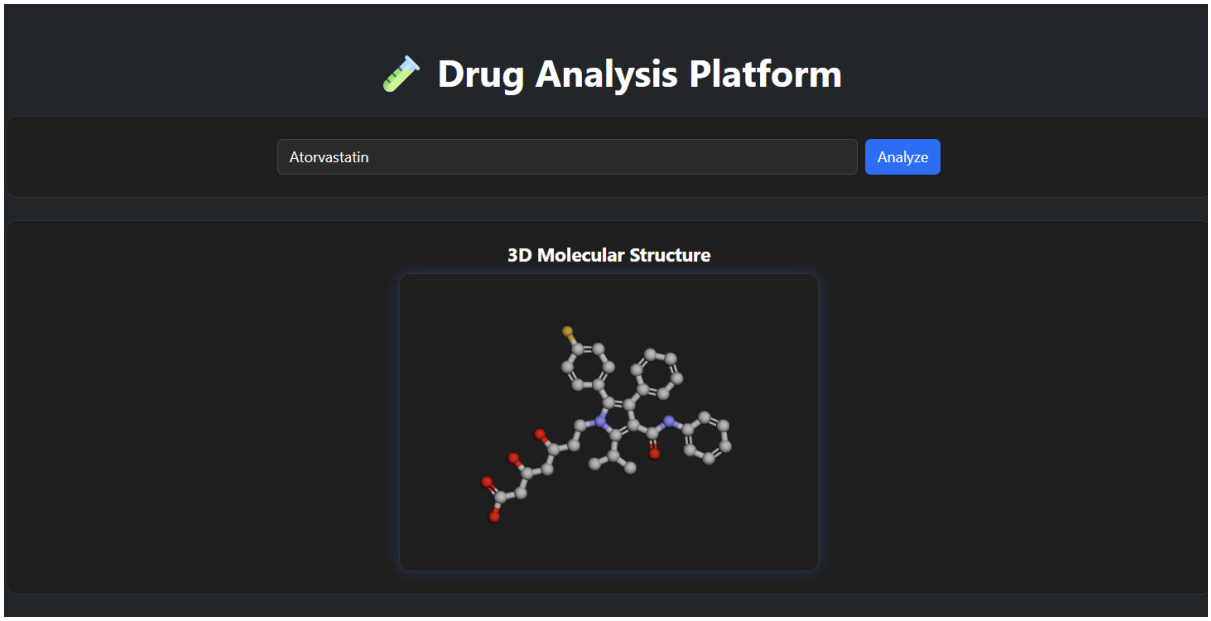


Figure 2: 3D Molecular Structure of Atorvastatin

RESULTS AND DISCUSSION:

The molecular descriptors of the selected drugs were computed using the developed web-based platform and validated using PubChem and SwissADME databases. The key physicochemical properties, including molecular weight (MW), lipophilicity (LogP), hydrogen bond donors (HBD), hydrogen bond acceptors (HBA), and topological polar surface area (TPSA), were analysed.

Table 1 summarises the computed molecular descriptors for Ibuprofen, Aspirin, and Paracetamol.

DRUG	MW	LogP	HBD	HBA	TPSA	Lipinski
Ibuprofen	206.28	3.07	1	1	37.3	Pass
Aspirin	180.16	1.31	1	3	63.6	Pass
Paracetamol	151.16	1.35	2	2	49.33	Pass

Table 1: Molecular Descriptor Comparison of Selected Drugs

All three drugs met Lipinski's Rule of Five, which shows they have good drug-like properties. However, we noticed significant differences in their physicochemical characteristics.

Ibuprofen had higher lipophilicity (LogP = 3.07) and lower TPSA, suggesting it can easily pass through membranes and is more hydrophobic. In contrast, Aspirin had higher polarity with increased TPSA, which indicates stronger hydrogen bonding potential and less permeability. Paracetamol showed moderate properties, balancing polarity and lipophilicity.

These differences in molecular descriptors are important in deciding how drugs behave, how they are absorbed, and their potential interactions.

Comparison with PubChem and SwissADME Databases

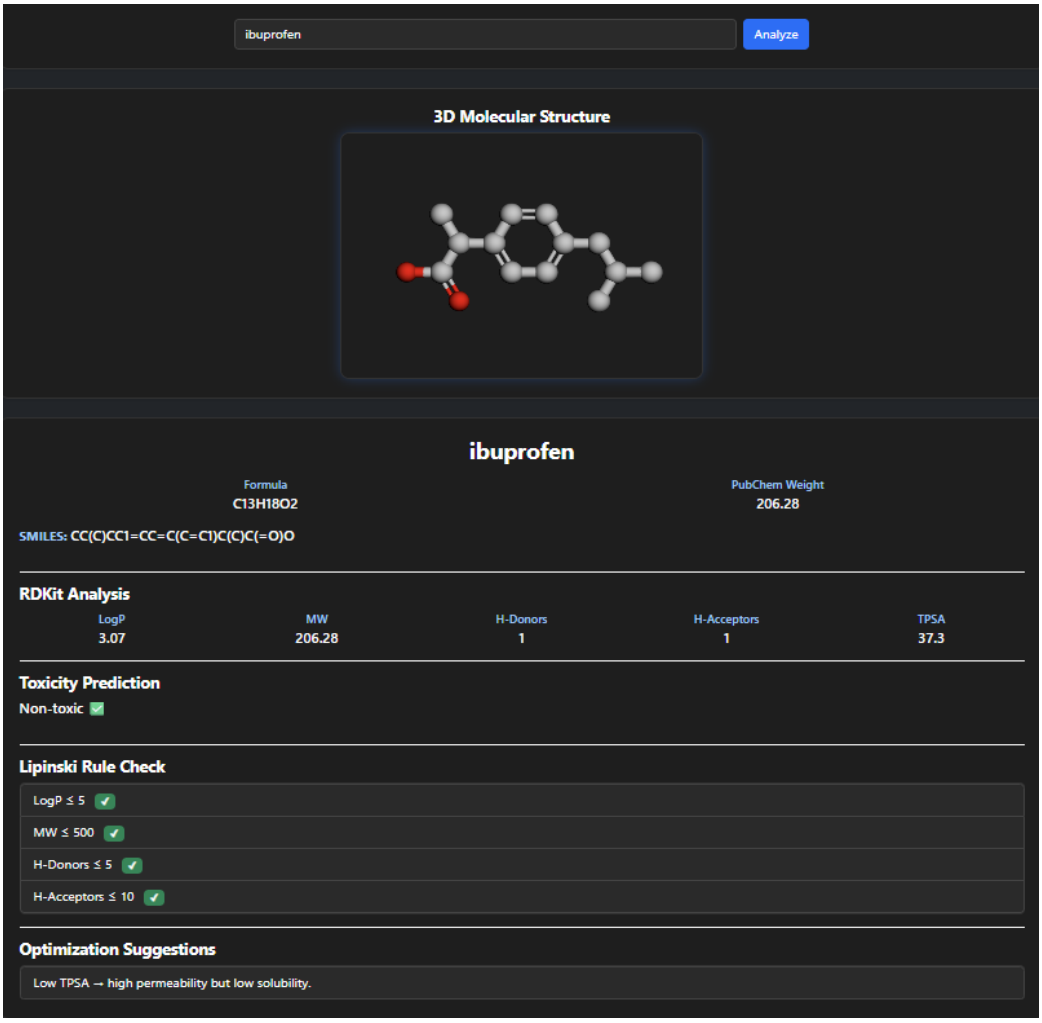
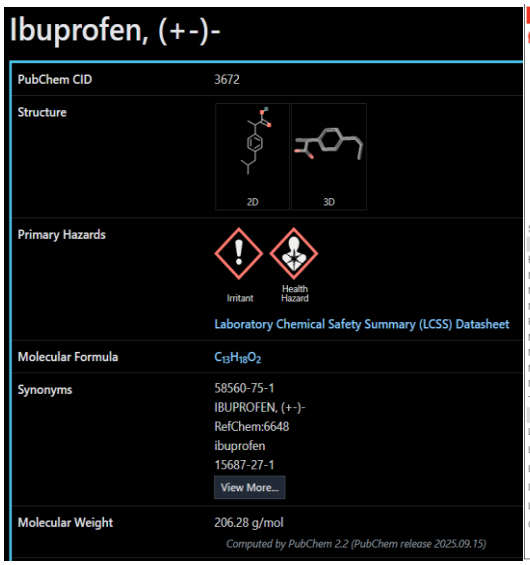


Figure 3: Developed web-based drug analysis platform for **Ibuprofen**



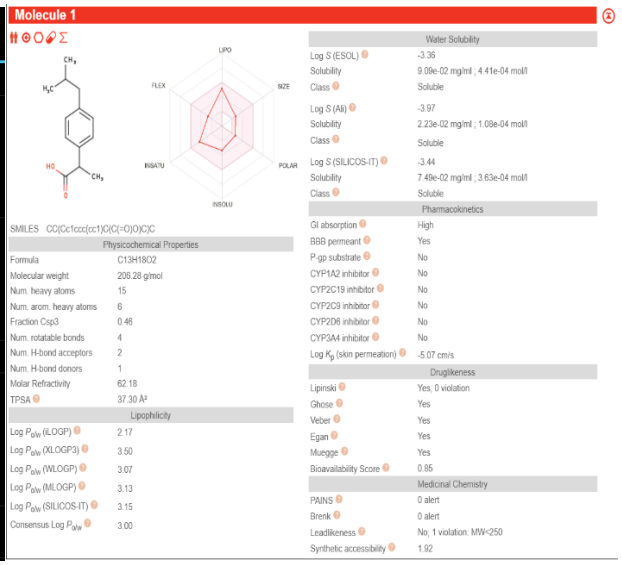


Figure 4: Mol data-PubChem database(**Ibuprofen**) **Figure 5:** Molecular descriptor – SwissADME

Comparison with PubChem and SwissADME Databases

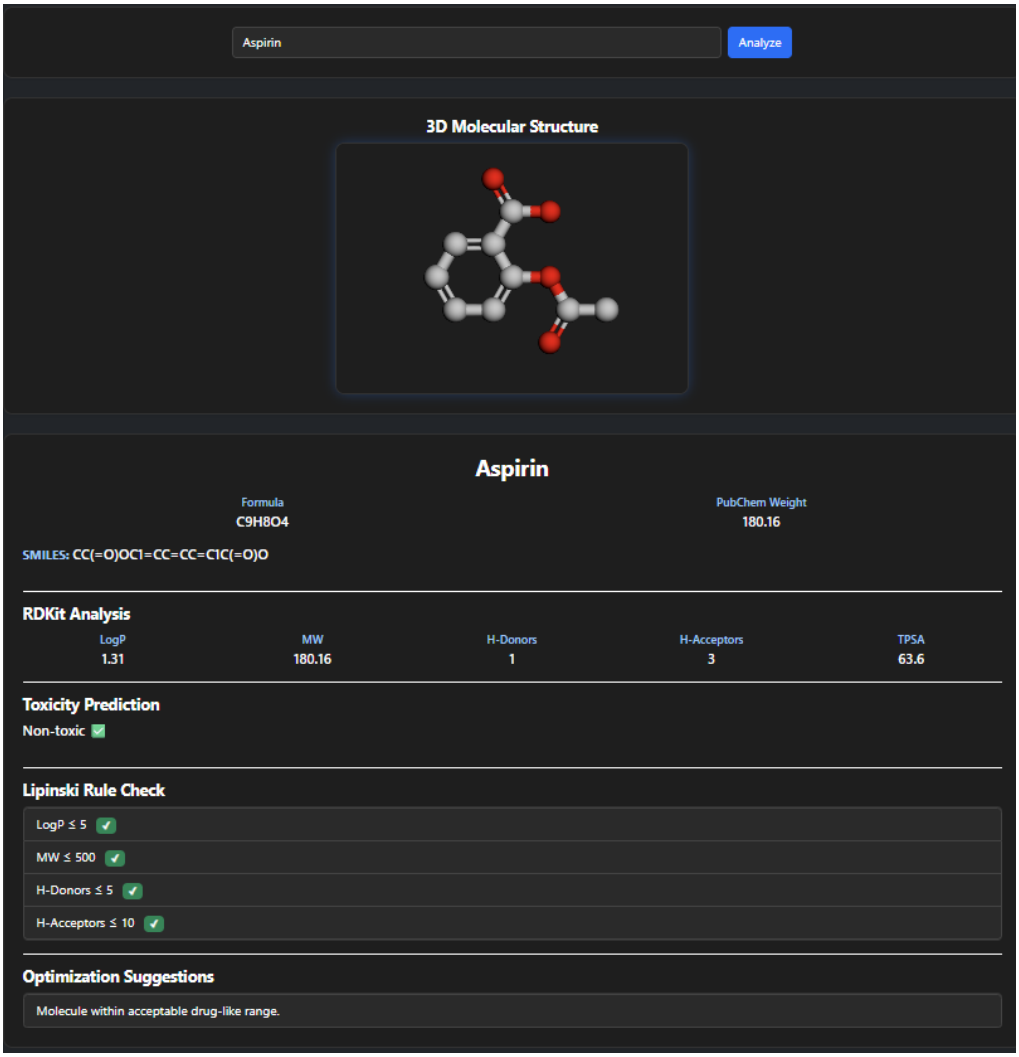
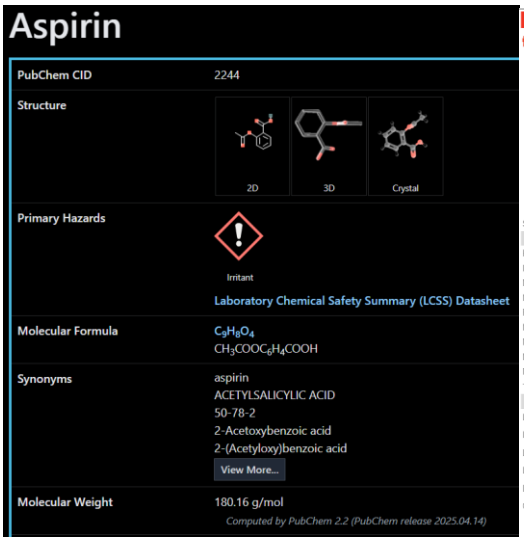


Figure 6: Developed web-based drug analysis platform for **Aspirin**



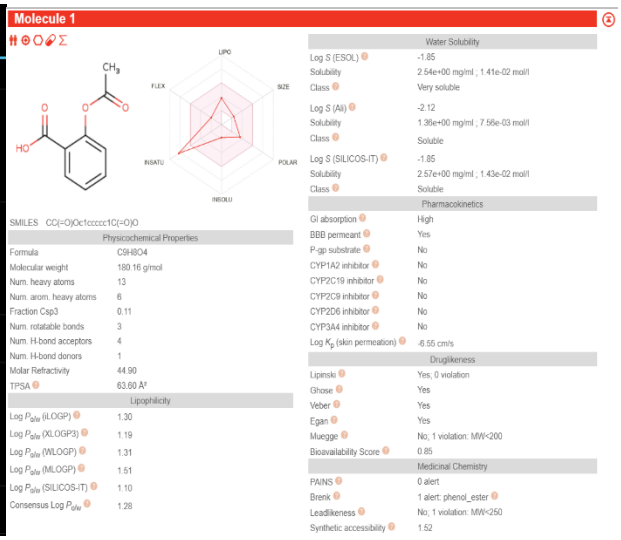


Figure 7: Mol data-PubChem database(**Aspirin**) **Figure 8:** Molecular descriptor – SwissADME(**Aspirin**)

Comparison with PubChem and SwissADME Databases

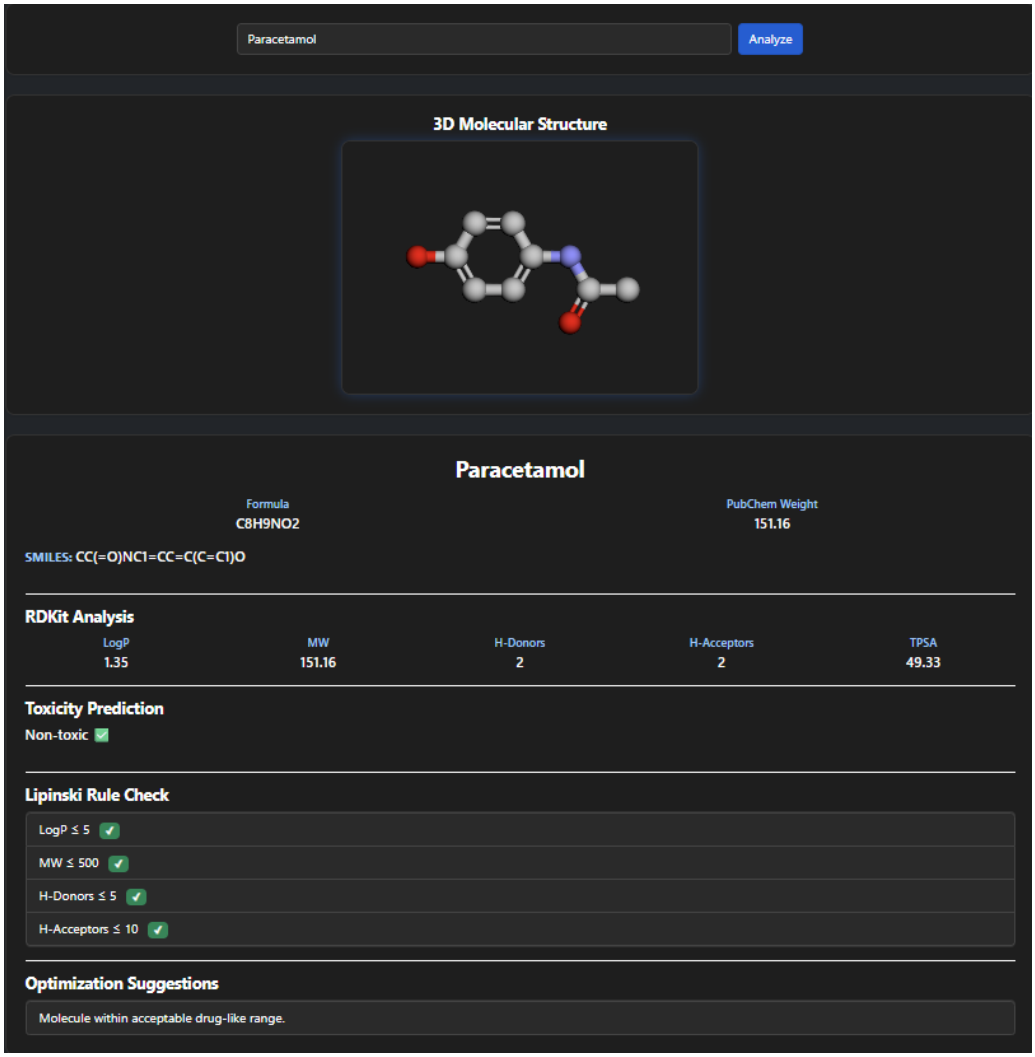
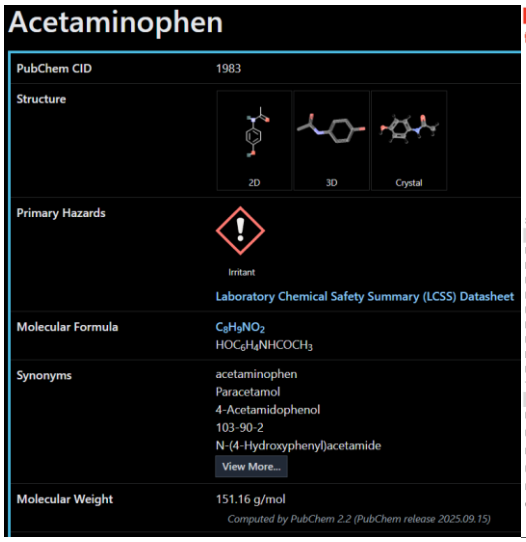


Figure 9: Developed web-based drug analysis platform for **Paracetamol.**



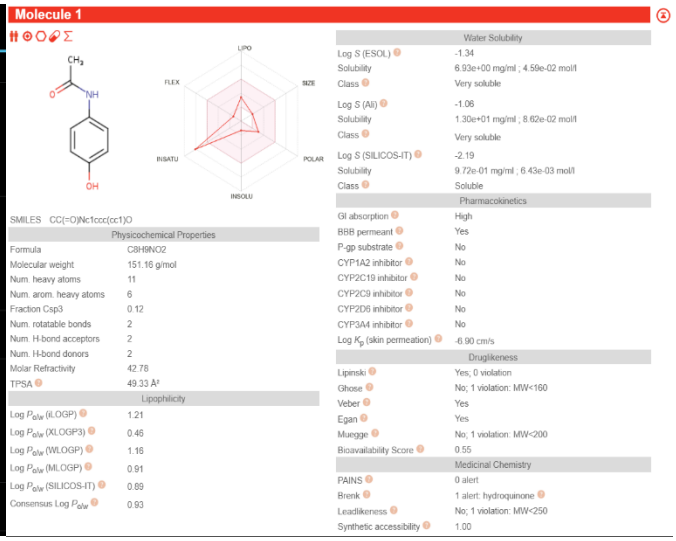


Figure 10: Mol data-PubChem database

Figure 11: Molecular descriptor – SwissADME

DRUG INTERACTION ANALYSIS:

Drug–drug interaction analysis was performed by comparing molecular properties and pharmacological mechanisms of the selected drugs.

The combination of Ibuprofen and Aspirin represents a high interaction risk scenario. Both drugs belong to the non-steroidal anti-inflammatory drug (NSAID) class and act on the same cyclooxygenase (COX) enzyme. Their similar molecular properties and overlapping mechanisms result in competitive binding and additive inhibition, increasing the risk of gastrointestinal toxicity.

In contrast, the combination of Paracetamol and Ibuprofen represents a lower interaction risk. Although both drugs are used for pain relief, their mechanisms of action differ. Paracetamol primarily acts centrally, whereas Ibuprofen acts peripherally through COX inhibition. Additionally, their physicochemical properties differ, reducing the likelihood of direct competition.

This comparison demonstrates that drug–drug interactions are influenced not only by molecular descriptors but also by pharmacological targets and mechanisms of action.

CONCLUSION:

This study successfully developed a web-based cheminformatics platform for evaluating drug-likeness and analyzing drug–drug interactions. The platform was able to compute key molecular descriptors, including molecular weight, LogP, hydrogen bond donors and acceptors, and topological polar surface area, and apply Lipinski's Rule of Five to assess drug-likeness.

The analysis of selected drugs demonstrated that variations in physicochemical properties significantly influence drug behavior and interaction potential. The combination of Ibuprofen and Aspirin showed higher interaction risk due to similar mechanisms and overlapping properties, whereas Paracetamol and Ibuprofen exhibited lower interaction potential due to differences in their mechanisms of action and molecular characteristics.

The results obtained from the developed platform were consistent with established databases such as PubChem and SwissADME, confirming the accuracy and reliability of the computational approach. This work highlights the importance of integrating cheminformatics tools for preliminary drug analysis and interaction prediction in medicinal chemistry.

Overall, the developed system provides a simple and effective framework for understanding drug properties and evaluating potential interactions using computational methods.

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